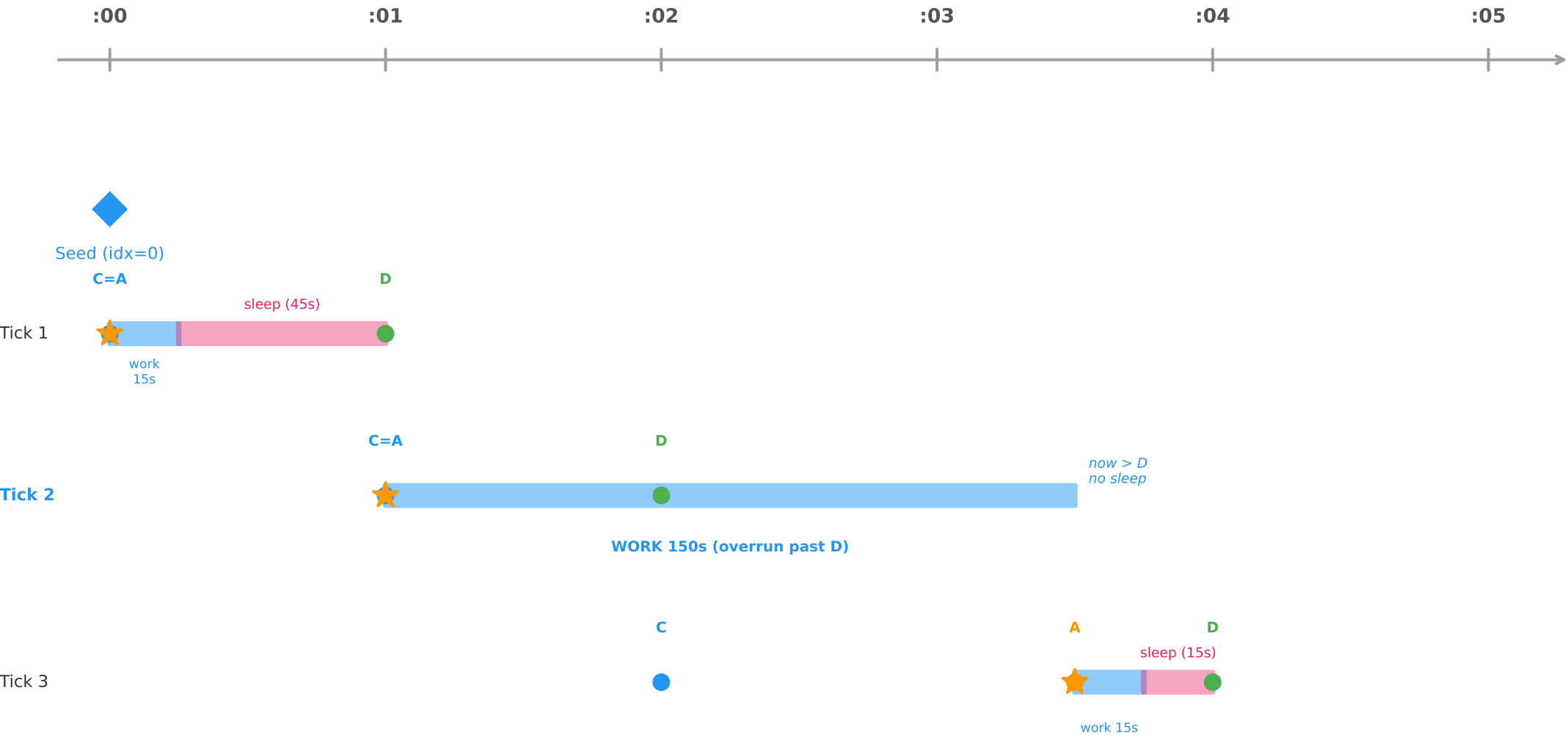


tickFuture - crontab(\_minutely)  
"emit first, then sleep remaining gap"



**emit = acquires (downstream receives the tick BEFORE sleeping)**

*Overrun skips the sleep, not the tick. `cron.next(:03:30) = :04`.*

Note: snooze (conclude - acquires) is the planned gap. Actual sleep =  $\max(0, \text{conclude} - \text{now\_after\_work})$ .

● C (commence)      ★ A (acquires/emit)      ● D (conclude)      sleep      work

**Tick Values at Emit Time**

|        | index | commence | acquires | conclude | snooze | work           |
|--------|-------|----------|----------|----------|--------|----------------|
| Tick 1 | 1     | :00:00   | :00:00   | :01:00   | 60s    | 15s            |
| Tick 2 | 2     | :01:00   | :01:00   | :02:00   | 60s    | 150s (overrun) |
| Tick 3 | 3     | :02:00   | :03:30   | :04:00   | 30s    | 15s            |